

Tutorial 42

Raiders, hedges, and the taxpayer's put — and your final package

Advanced Corporate Finance (EBM232A05)

University of Groningen

- **Problem set 42, problems 1–4** — the free-rider (Week 41) and hedging plus the deposit-insurance put (Week 42), at the board.
- **Problems 5–6 are self-study**: they need Tuesday's final lecture (payout), and next week's tutorial is the code defence. Key on Brightspace; office hours for questions.
- Last fifteen minutes: the **M3 checklist** — your final package is due at the end of this week.

Problem set 42: P1–P4

A raider can lift value per share from v to $v' > v$ at cost c ; he bids p , conditional on control. Holders are small — each takes success as given.

- (a) The one holder's payoff table, and the tender condition on p .
- (b) The raider's profit; where does $v' - v$ go?
- (c) Why is no rival bidder needed? What exactly is underprovided, and who free-rides on whom?
- (d) Acquirer returns ≈ 0 is also what bidder competition predicts. One observable that separates the stories.

Not pivotal $\Rightarrow$ tender only if $p \geq v'$, so

$$\pi = (v' - p) - c \leq -c < 0 \quad \Rightarrow \quad \text{no bid.}$$

- The whole improvement goes to the *selling* shareholders — and is therefore **never created**. The improvement is a public good; every small holder free-rides on the raider's costly work.
- No competition anywhere in the argument: a **monopolist** raider earns zero too.
- Separators (Moeller et al., 2005): rival bids are rare ($\approx 1.4\%$); and under free-riding the gain is often **never attempted** — announcement returns cannot settle whether takeovers create value.

$v = 40$, $v' = 50$, $c = 2$. Toehold α : bought quietly at v . Dilution φ : holders tender at $v' - \varphi$; the raider keeps φ per share bought.

- (a) Neither instrument: bid? Toehold only: minimum α — and why disclosure at 3–5% makes it academic.
- (b) Dilution only ($\alpha = 0$): φ^* ?
- (c) A *bad* raider ($v'' = 35$, same c), still $\alpha = 0$: his φ^{**} ? Compare.
- (d) Both raiders at $\alpha = 10\%$: what does the toehold do to the two thresholds?

- (a) $\pi = -2$: no bid on a 25% improvement. Toehold: $\alpha \geq c/(v' - v) = \mathbf{20\%}$ — but disclosure rules cap quiet stakes at 3–5%, so the fix is real and unavailable.
- (b) $\pi = \varphi - c$: $\varphi^* = \mathbf{2}$.
- (c) The bad raider's π is *identical*: $\varphi^{**} = \mathbf{2} = \varphi^*$. **At $\alpha = 0$ dilution is blind** — it pays for control, not improvement; any charter that admits discipline admits plunder at the same moment.
- (d) At $\alpha = 10\%$: $\varphi^* = 1/0.9 \approx \mathbf{1.11}$, $\varphi^{**} = 2.5/0.9 \approx \mathbf{2.78}$. Only the toehold sees $v' - v$: it pays the good raider, taxes the bad one, and opens a band $(1.11, 2.78)$ that buys discipline without looting.

$w \in \{30, 130\}$ equally likely; $I = 80$, $G = 30$, outside finance at $\kappa = 0.4$;
 $f(w) = w + G - \kappa(I - w)^+$. A hedge replaces the two states by their mean.

- (a) f in both states; unhedged expectation; hedged value; the gain.
- (b) Show the gain = probability $\times$ deadweight cost avoided. Which parameter switches it off?
- (c) Shareholders are diversified. Why hedge anyway — what replaces risk aversion, and which way does the curvature go?
- (d) Let G be high exactly when w is high ($\rho > 0$). Still hedge fully? What does this make of Tufano (1996)'s barely-hedging gold miners?

P3 — solved

$f(30) = 40$, $f(130) = 160$, $\mathbb{E}[f] = 100$; hedged: $f(80) = \mathbf{110}$. Gain = $\mathbf{10} = \frac{1}{2} \times 20$ — half the bad state's deadweight financing cost.

- $\kappa = 0$: f linear, chord on the curve, hedging worthless — **MM**. The friction is the same κ as PS37 and Week 38.
- Risk aversion is replaced by **technology**: costly outside finance makes f *concave*, so $f(\mathbb{E}[w]) > \mathbb{E}[f(w)]$. Week 39 was the mirror image — convex claim, chord *above* the curve, variance *wanted*.
- $\rho > 0$: cash already arrives when there is something to spend it on — hedge *partially*. Gold miners are the textbook $\rho > 0$ case, so their thin hedging is **consistent with** FSS, not a refutation.

Assets: expected 100; insured deposits $D = 90$, capital $w = 10$. Two equal-mean books: **safer** {75, 125}, **riskier** {55, 145}.

- (a) Expected equity and insurer cost S for each book. Who pays for the shift?
- (b) Which option does each party hold? Sign $\partial S / \partial \sigma$ without computing.
- (c) Repeat at $w = 25$ and $w = 50$. Where does the incentive die, and why is it a *threshold*?
- (d) “Deposit insurance caused the risk it insures” — true in which sense, false in which, and what is capital regulation in the model’s language?

P4 — solved

Capital	Safer: equity, S	Riskier: equity, S	Shift: equity	insurer
$w = 10$	17.5, 7.5	27.5, 17.5	+10	+10
$w = 25$	25, 0	35, 10	+10	+10
$w = 50$	50, 0	50, 0	0	0

- Equity = **call** struck at D ; the insurer **wrote a put**: $\partial S / \partial \sigma > 0$ — Week 39's vega, held by the taxpayer.
- The incentive dies only at $w \geq 45$: a **threshold** — while equity is wiped out in the bad state, deepening the wipe-out is free. **Tripling capital from 10 to 25 changed nothing.**
- Capital regulation: **a minimum deductible on the put**, set so shareholders live in the bad state again; half-measures are worthless.

P5–P6 — self-study, after Tuesday's lecture

- P5 works the Week-43 payout model by hand: the manager's stake W_t as a running score, payout at the ceiling, liquidation at the floor — and why two firms with identical earnings pay differently.
- P6 reads the payout evidence (Lintner; Leary–Michaelis; Baker–Wurgler) and closes with the complete frictions ledger, all seven weeks — your revision sheet for the exam.
- Key on Brightspace with the set. Office hours for questions — next week's tutorial is the code defence, so there is no discussion slot.

M3 — the final package

The M3 checklist (due end of this week)

Run it **before** you submit — the grading harness runs it after:

- `run_all` regenerates your submitted numbers **in a clean environment** — the anti-hardcoding check. Hardcoded tables earn nothing.
- TIER structure; README with the data-availability statement; environment file; **no raw WRDS data anywhere in the history**.
- Dossier complete: verbatim transcripts, underspecification catalogue (≥ 3), catches (≥ 3 , each citing a transcript line *and* a fixing commit), the fresh-context Referee-2 audit, the disclosure statement.
- Extension on your **assigned axis** (a different axis without agreement caps D4 at 10).
- Commit history that shows the work happening — it is graded, and it defends you next week.

Before you leave

- **M3 due end of this week** — exact time on Brightspace. No extensions into defence week.
- **Next week:** Tuesday, the final lecture (payout). Wednesday, the **code defence** — every team member must be able to walk through every part of the repo. Individual reflections (D3b) due at the start of the week.
- **Problem set 42 key** on Brightspace after today; P5–P6 after Tuesday's lecture.

- Moeller, S. B., F. P. Schlingemann, and R. M. Stulz (2005). **“Wealth Destruction on a Massive Scale? A Study of Acquiring-Firm Returns in the Recent Merger Wave”**. In: *The Journal of Finance* 60.2, pp. 757–782.
- Tufano, P. (1996). **“Who Manages Risk? An Empirical Examination of Risk Management Practices in the Gold Mining Industry”**. In: *The Journal of Finance* 51.4, pp. 1097–1137.